

Name : Sidhish p

RegNo: 12218130

Create:

1.Insert a new student into the collection with your own details.

```
colleges> db.students.insertOne( {  
...   "name": "sidhish",  
...   "rollNo": "CSE005",  
...   "branch": "CSE",  
...   "year": 2,  
...   "email": "sidhish.s@example.com",  
...   "marks": {  
...     "maths": 98,  
...     "physics": 90,  
...     "chemistry": 972  
...   },  
...   "isHosteller": true  
... })  
{  
  acknowledged: true,  
  insertedId: ObjectId('685e56a0323518e287748a67')  
}
```

2.Add 3 students in one command using insertMany.

```
colleges> db.students.insertMany( [{  
...   "name": "jagga",  
...   "rollNo": "CSE006",  
...   "branch": "CSE",  
...   "year": 3,  
...   "email": "jagga.k@example.com",  
...   "marks": {  
...     "maths": 95,  
...     "physics": 82,  
...     "chemistry": 62  
...   },  
...   "isHosteller": false  
... },  
... {  
...   "name": "sriram",  
...   "rollNo": "CSE007",  
...   "branch": "CSE",  
...   "year": 2,  
...   "email": "sriram.a@example.com",  
...   "marks": {  
...     "maths": 78,  
...     "physics": 50,  
...     "chemistry": 72  
...   },  
...   "isHosteller": true  
... },  
... {  
...   "name": "seshu",  
...   "rollNo": "CSE005",  
...   "branch": "CSE",  
...   "year": 4,  
...   "email": "seshu.k@example.com",  
...   "marks": {  
...     "maths": 60,  
...     "physics": 78,  
...     "chemistry": 57  
...   },  
...   "isHosteller": true  
... }  
... ]  
... )  
{  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('685e58c1323518e287748a68'),  
    '1': ObjectId('685e58c1323518e287748a69'),  
    '2': ObjectId('685e58c1323518e287748a6a')  
  }  
}
```

2.

Read:

3. Find all students in the CSE branch.

```
colleges> db.students.find({branch:"CSE"})
[
  {
    _id: ObjectId('685e5573323518e287748a63'),
    name: 'Amit Sharma',
    rollNo: 'CSE001',
    branch: 'CSE',
    year: 2,
    email: 'amit.sharma@example.com',
    marks: { maths: 78, physics: 85, chemistry: 72 },
    isHosteller: true
  },
  {
    _id: ObjectId('685e5573323518e287748a66'),
    name: 'Anjali Singh',
    rollNo: 'CSE004',
    branch: 'CSE',
    year: 4,
    email: 'anjali.singh@example.com',
    marks: { maths: 91, physics: 92, chemistry: 89 },
    isHosteller: false
  },
  {
    _id: ObjectId('685e56a0323518e287748a67'),
    name: 'sidhish',
    rollNo: 'CSE005',
    branch: 'CSE',
    year: 2,
    email: 'sidhish.s@example.com',
    marks: { maths: 98, physics: 96, chemistry: 97 },
    isHosteller: true
  },
  {
    _id: ObjectId('685e58c1323518e287748a68'),
    name: 'jagga',
    rollNo: 'CSE006',
    branch: 'CSE',
    year: 3,
    email: 'jagga.k@example.com',
    marks: { maths: 95, physics: 82, chemistry: 62 },
    isHosteller: false
  },
  {
    _id: ObjectId('685e58c1323518e287748a69'),
    name: 'sriram',
    rollNo: 'CSE007',
    branch: 'CSE',
    year: 2,
    email: 'sriram.a@example.com',
    marks: { maths: 78, physics: 50, chemistry: 72 },
    isHosteller: true
  },
  {
    _id: ObjectId('685e58c1323518e287748a6a'),
    name: 'seshu',
    rollNo: 'CSE005',
    branch: 'CSE',
    year: 4,
    email: 'seshu.k@example.com',
    marks: { maths: 60, physics: 78, chemistry: 57 },
    isHosteller: true
  }
]
colleges> [{
```

4. Find students who are in 2nd year and are hostellers.

```
colleges> db.students.find({year:2},{isHosteler:true})
[
  { _id: ObjectId('685e5573323518e287748a63') },
  { _id: ObjectId('685e56a0323518e287748a67') },
  { _id: ObjectId('685e58c1323518e287748a69') }
]
colleges> db.students.find({marks:{maths:{$gt:90}}})
```

5. Find the student who scored more than 90 in Maths.

```
colleges> db.students.find({"marks.maths":{"$gt:90}})
[
  {
    _id: ObjectId('685e5573323518e287748a66'),
    name: 'Anjali Singh',
    rollNo: 'CSE004',
    branch: 'CSE',
    year: 4,
    email: 'anjali.singh@example.com',
    marks: { maths: 91, physics: 92, chemistry: 89 },
    isHosteller: false
  },
  {
    _id: ObjectId('685e56a0323518e287748a67'),
    name: 'sidhish',
    rollNo: 'CSE005',
    branch: 'CSE',
    year: 2,
    email: 'sidhish.s@example.com',
    marks: { maths: 98, physics: 90, chemistry: 972 },
    isHosteller: true
  },
  {
    _id: ObjectId('685e58c1323518e287748a68'),
    name: 'jagga',
    rollNo: 'CSE006',
    branch: 'CSE',
    year: 3,
    email: 'jagga.k@example.com',
    marks: { maths: 95, physics: 82, chemistry: 62 },
    isHosteller: false
  }
]
colleges>
```

6. Show name and branch of all students (only those fields).

```
colleges> db.students.find({}, {name:1, branch:1, _id:0})
[
  { name: 'Amit Sharma', branch: 'CSE' },
  { name: 'Pooja Verma', branch: 'ECE' },
  { name: 'Rahul Kumar', branch: 'ME' },
  { name: 'Anjali Singh', branch: 'CSE' },
  { name: 'sidhish', branch: 'CSE' },
  { name: 'jagga', branch: 'CSE' },
  { name: 'sriram', branch: 'CSE' },
  { name: 'seshu', branch: 'CSE' }
]
colleges>
```

7. Count the number of students in CSE branch.

```
colleges> db.students.find({branch:"CSE"}).count()
6
colleges> |
```

8. Get all students sorted by their Physics marks in descending order.

```
6 colleges> db.students.find().sort({"marks.physics":-1})
[
  {
    _id: ObjectId('685e5573323518e287748a66'),
    name: 'Anjali Singh',
    rollNo: 'CSE004',
    branch: 'CSE',
    year: 4,
    email: 'anjali.singh@example.com',
    marks: { maths: 91, physics: 92, chemistry: 89 },
    isHosteller: false
  },
  {
    _id: ObjectId('685e56a0323518e287748a67'),
    name: 'Sidhish',
    rollNo: 'CSE005',
    branch: 'CSE',
    year: 2,
    email: 'sidhish.s@example.com',
    marks: { maths: 98, physics: 90, chemistry: 972 },
    isHosteller: true
  },
  {
    _id: ObjectId('685e5573323518e287748a63'),
    name: 'Amit Sharma',
    rollNo: 'CSE001',
    branch: 'CSE',
    year: 2,
    email: 'amit.sharma@example.com',
    marks: { maths: 78, physics: 85, chemistry: 72 },
    isHosteller: true
  },
  {
    _id: ObjectId('685e58c1323518e287748a68'),
    name: 'Jagga',
    rollNo: 'CSE006',
    branch: 'CSE',
    year: 3,
    email: 'jagga.k@example.com',
    marks: { maths: 95, physics: 82, chemistry: 62 },
    isHosteller: false
  },
  {
    _id: ObjectId('685e5573323518e287748a65'),
    name: 'Rahul Kumar',
    rollNo: 'ME003',
    branch: 'ME',
    year: 1,
    email: 'rahul.kumar@example.com',
    marks: { maths: 82, physics: 79, chemistry: 88 },
    isHosteller: true
  },
  {
    _id: ObjectId('685e58c1323518e287748a6a'),
    name: 'Sesha',
    rollNo: 'CSE005',
    branch: 'CSE',
    year: 4,
    email: 'sesha.k@example.com',
    marks: { maths: 60, physics: 78, chemistry: 57 },
    isHosteller: true
  },
  {
    _id: ObjectId('685e5573323518e287748a64'),
    name: 'Pooja Verma',
    rollNo: 'ECE002',
    branch: 'ECE',
    year: 3,
    email: 'pooja.verma@example.com',
    marks: { maths: 67, physics: 74, chemistry: 69 },
    isHosteller: false
  },
  {
    _id: ObjectId('685e58c1323518e287748a69'),
    name: 'Sriram',
    rollNo: 'CSE007',
    branch: 'CSE',
    year: 2,

```

Update:

9. Update the Chemistry marks of roll number "CSE001" to 80.

```
colleges> db.students.updateOne({rollNo:"CSE001"},{$set:{"marks.chemistry":80}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
colleges>
```

10. Add a new field CGPA with default value 8.0 to all students.

```
colleges> db.students.updateMany({}, {$set:{CGPA:"8.0"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 8,
  modifiedCount: 8,
  upsertedCount: 0
}
colleges>
```

11. Change hosteller status to false for all 4th-year students.

```
colleges> db.students.updateMany({year:4}, {$set:{isHosteller:false}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 1,
  upsertedCount: 0
}
colleges>
```

12. Rename the field branch to department.

```
colleges> db.students.updateMany({}, {$rename:{"branch":"department"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 8,
  modifiedCount: 8,
  upsertedCount: 0
}
colleges>
```

Delete:

13. Delete the student whose roll number is "ECE002".

```
colleges> db.students.deleteOne({rollNo:"ECE002"})  
{ acknowledged: true, deletedCount: 1 }  
colleges>
```

14. Delete all students who are in 1st year.

```
colleges> db.students.deleteMany({year:1})  
{ acknowledged: true, deletedCount: 1 }  
colleges>
```